

compiler that you really want to do this dumb thing.

A Superclass object is not a Subclass object. An HourlyEmployee is an Employee (Subclass is a Superclass object) but an Employee is *not* an HourlyEmployee (Superclass is *not* a Subclass object). If you want to actually call some of the methods that don't exist in the Superclass, then you have to first cast the object back to a reference of its own kind.

5.5.1 Constructors and finalisers in subclasses

Whenever you instantiate a Subclass object, the Constructors for the Superclass should be called to initialise the instance variables of the Superclass. This instantiation can be done either implicitly or explicitly (via the super reference). These calls to instantiate the Superclass object must be the first thing in your class's Constructor method. The opposite of the Constructor method is the finaliser method. If your class bothers to define a finalizer method, then the implicit or explicit call to the Superclass finaliser must be the last thing in the finalizer method. You will always want to define your finaliser method as protected, so that Subclasses that inherit from your class can access its finalizer. Classes that only use Superclass objects (not inherit them) cannot use the Superclass's finalizer methods.

You must always use this "super" reference when you intend to call the finaliser of the Superclass. If you omit the super class, you're actually calling the finaliser of the method you're already inside of – which is called infinite recursion. You cannot do a cascading call to a Superclass. Instead, you have to rely on the Superclass itself to call its own Superclass.

5.5.2 Implicit Subclass-object to Superclass-object conversion

In order to understand the concept of subclass and superclass object, let's consider the following illustration. Ravi was feeling entrepreneurial, and so he decided to buy a piece of land in Bihar. Ravi paid ₹50 for what the deed described as a wooden shack in the swamp. But, to his great surprise, when he got there, Ravi discovered he hadn't bought a shack in a swamp, but rather, the Taj Mahal on a beautiful lake. Ravi was ecstatic. Ravi got more than he was promised. Maya was feeling entrepreneurial too, and so she decided to buy a huge mansion on an estate in Bihar. She paid ₹5 Million for what the deed described as a palatial estate on 40 acres of land. But, to her great surprise, found she hadn't bought a magnificent mansion and estate, but rather, a Doublewide trailer next to a dump. Maya was horrified. Maya got less than was promised.